

Sharwil Bhende

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OBJECTIVE

Computer Engineering student (B.Tech, Class of 2027) at Symbiosis Institute of Technology seeking an AI/ML Internship to apply hands-on experience in machine learning, deep learning, computer vision, and LLM-based agentic systems to real-world problems. Track record of shipping end-to-end AI applications with Python and modern ML frameworks, backed by two conference publications and a national AI hackathon build.

SKILLS

AI/ML: Machine Learning, Deep Learning, Computer Vision, Generative AI, Agentic AI

LLM/AI: Gemini API, Function Calling, Retrieval-Augmented Generation (RAG)

Frameworks & Tools: MediaPipe, YOLOv8, FastAPI, Streamlit

Languages: Python, C/C++, SQL

Core CS: Data Structures & Algorithms, DBMS, Object-Oriented Programming (OOP), Operating Systems

EDUCATION

Symbiosis Institute of Technology

B.Tech, Computer Engineering

Nagpur, India

2023 – 2027

PROJECTS

AI Support Intelligence: RAG-Based Query Classification & Routing

- Built an end-to-end support intelligence system on the BANKING77 dataset (77 intents), benchmarking a TF-IDF + Logistic Regression baseline against a fine-tuned DistilBERT classifier.
- Designed a Retrieval-Augmented Generation (RAG) pipeline using FAISS vector search and Gemini for grounded responses, with confidence-based routing that escalates low-confidence queries from the classifier to the LLM.
- Shipped a FastAPI backend and a React dashboard that surfaces predictions, confidence scores, and routing decisions for each query.

Real-Time Indian Sign Language Recognition

- Building a real-time ML pipeline for Indian Sign Language recognition, extracting pose, hand, and face landmarks with MediaPipe Holistic and classifying gesture sequences with a GRU-based temporal model.
- Architecting an agentic fallback layer that invokes Gemini function-calling on low-confidence predictions to assemble coherent sentences.
- Collecting a real, signer-independent dataset with a partner organization to support a research manuscript in preparation.

Failed-Subscription Revenue Recovery Agent

- Built an agentic AI system for the Razorpay AI Buildathon 2026 that diagnoses failed UPI/card subscription payments and autonomously selects recovery actions using Gemini function-calling.
- Designed pay-cycle-aware retries and escalating multi-channel notifications under a safety layer that overrides low-confidence AI decisions, keeping every action bounded and auditable.
- Shipped a live Streamlit dashboard with batch metrics, per-case audit trails, and reasoning logs; the agent outperformed a naive retry baseline on a 140-case synthetic evaluation (75.7% vs. 57.1% recovery).

Autoregent: Self-Healing API Gateway

- Built a FastAPI gateway using Gemini to diagnose and remediate schema drift live while enforcing AI-assisted healing only on eligible read-path routes.
- Implemented fail-loud guards including confidence thresholds, timeouts, unrecoverable-drift detection, loop detection, per-route circuit breakers, and heal budgets.
- Added HMAC-signed disclosure events and a real-time Next.js dashboard; deployed the gateway, dashboard, and signed event log on Railway.

RESEARCH & PUBLICATIONS

- Presented “Enhancing Cotton Yield Through Detection of Cotton Leaf Curl Disease” at IC3I Conference.
- Presented “Comparative Study of YOLOv5, YOLOv8, Faster R-CNN, SSD for Traffic Detection” at IICTDS 2025.
- Deep learning-based Indian Sign Language detection; manuscript in preparation for journal submission.

EXTRACURRICULAR

- Secretary-General, Model United Nations.** Led and coordinated an event of 100+ participants.
- Represented as a delegate across multiple committees, developing strong negotiation, diplomacy, and public speaking skills.

ACHIEVEMENTS

- Shri C. Subramaniam Award for Excellence in Character, Bharatiya Vidya Bhavan.** Recognized for leadership, integrity, and strong personal values.
- Top Performer, Industry Readiness Program 2025–2026,** Symbiosis Institute of Technology, Nagpur.